



qualcomm/**PiperTTS-DE**



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Text-to-Audio

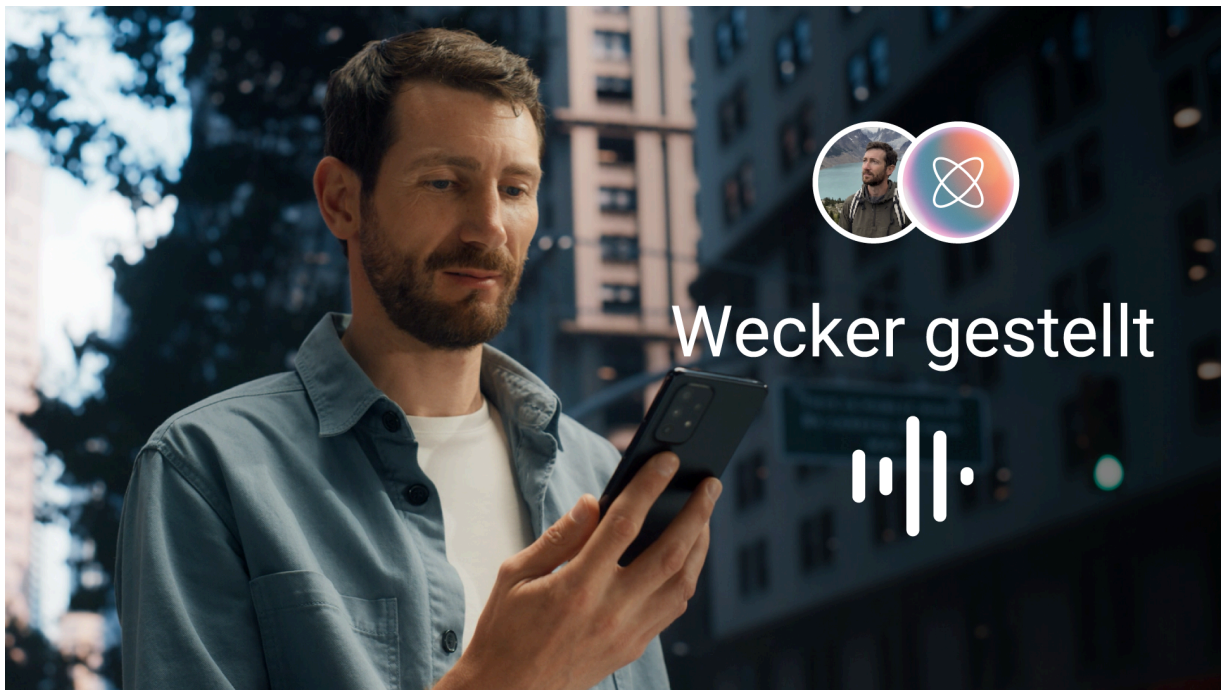
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## **PiperTTS-DE: Optimized for Qualcomm Devices**

PiperTTS is a high-quality multi-lingual text-to-speech library.

This is based on the implementation of PiperTTS-DE found [here](#). This repository contains pre-exported model files optimized for Qualcomm® devices. You can use the [Qualcomm® AI Hub Models](#) library to export with custom configurations. More details on model performance across various devices, can be found [here](#).

Qualcomm AI Hub Models uses [Qualcomm AI Hub Workbench](#) to compile, profile, and evaluate this model. [Sign up](#) to run these models on a hosted Qualcomm® device.

## **Getting Started**

There are two ways to deploy this model on your device:

### **Option 1: Download Pre-Exported Models**

Below are pre-exported model assets ready for deployment.

| Runtime  | Precision | Chipset                               | SDK Versions | Download                 |
|----------|-----------|---------------------------------------|--------------|--------------------------|
| VOICE_AI | float     | Snapdragon® X2 Elite                  | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Snapdragon® X Elite                   | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Snapdragon® 8 Gen 3 Mobile            | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Snapdragon® 8 Gen 1 Mobile            | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Qualcomm® Dragonwing™ QCS8550 (Proxy) | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Qualcomm® SA8775P                     | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Qualcomm® Dragonwing™ IQ-9075         | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Qualcomm® SA7255P                     | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Qualcomm® SA8295P                     | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Snapdragon® 8 Elite Mobile            | QAI RT 2.45  | <a href="#">Download</a> |
| VOICE_AI | float     | Snapdragon® 8 Elite Gen 5 Mobile      | QAI RT 2.45  | <a href="#">Download</a> |

For more device-specific assets and performance metrics, visit [PiperTTS-DE on Qualcomm® AI Hub](#).

### Option 2: Export with Custom Configurations

Use the [Qualcomm® AI Hub Models](#) Python library to compile and export the model with your own:

- Custom weights (e.g., fine-tuned checkpoints)
- Custom input shapes
- Target device and runtime configurations

This option is ideal if you need to customize the model beyond the default configuration provided here.

See our repository for [PiperTTS-DE on GitHub](#) for usage instructions.

## Model Details

**Model Type:** Model\_use\_case.audio\_generation

### Model Stats:

- Model checkpoint: rhasspy/piper-checkpoints
- Max decoded sequence length: 64 tokens
- Number of parameters (encoder): 7.51M
- Model size (encoder) (float): 28.7 MB
- Number of parameters (sdp): 1.04K
- Model size (sdp) (float): 6.61 KB
- Number of parameters (flow): 7.39M
- Model size (flow) (float): 28.2 MB
- Number of parameters (decoder): 1.66M
- Model size (decoder) (float): 6.37 MB
- Number of parameters (t5\_encoder): 15.1M
- Model size (t5\_encoder) (float): 57.5 MB
- Number of parameters (t5\_decoder): 5.72M
- Model size (t5\_decoder) (float): 21.8 MB

## Performance Summary

| Model           | Runtime  | Precision | Chipset                               | Inference Time (ms) | Peak Memory Range (MB) | Primary Compute Unit |
|-----------------|----------|-----------|---------------------------------------|---------------------|------------------------|----------------------|
| charsiu_decoder | VOICE_AI | float     | Snapdragon® X2 Elite                  | 0.34 ms             | 1 - 1 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Snapdragon® X Elite                   | 0.411 ms            | 1 - 1 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Snapdragon® 8 Gen 3 Mobile            | 0.309 ms            | 0 - 8 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Snapdragon® 8 Gen 1 Mobile            | 0.573 ms            | 1 - 10 MB              | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® Dragonwing™ QCS8275         | 0.997 ms            | 0 - 9 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® Dragonwing™ QCS8550 (Proxy) | 0.399 ms            | 0 - 3 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® SA8775P                     | 0.653 ms            | 0 - 10 MB              | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® SA8650P                     | 0.653 ms            | 0 - 10 MB              | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® SA8255P                     | 0.653 ms            | 0 - 10 MB              | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® QCS8450                     | 0.573 ms            | 1 - 10 MB              | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® Dragonwing™                 | 0.52 ms             | 1 - 3 MB               | NPU                  |

| Model           | Runtime  | Precision | Chipset                          | Inference Time (ms) | Peak Memory Range (MB) | Primary Compute Unit |
|-----------------|----------|-----------|----------------------------------|---------------------|------------------------|----------------------|
|                 |          |           | IQ-9075                          |                     |                        |                      |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® Dragonwing™ IQ-X7181   | 0.411 ms            | 1 - 1 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® Dragonwing™ Q-8750     | 0.28 ms             | 0 - 13 MB              | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® SA7255P                | 0.997 ms            | 0 - 9 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Qualcomm® SA8295P                | 0.802 ms            | 0 - 6 MB               | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Snapdragon® 8 Elite Mobile       | 0.28 ms             | 0 - 13 MB              | NPU                  |
| charsiu_decoder | VOICE_AI | float     | Snapdragon® 8 Elite Gen 5 Mobile | 0.261 ms            | 0 - 10 MB              | NPU                  |
| charsiu_encoder | VOICE_AI | float     | Snapdragon® X2 Elite             | 0.67 ms             | 0 - 0 MB               | NPU                  |
| charsiu_encoder | VOICE_AI | float     | Snapdragon® X Elite              | 1.062 ms            | 0 - 0 MB               | NPU                  |
| charsiu_encoder | VOICE_AI | float     | Snapdragon® 8 Gen 3 Mobile       | 0.637 ms            | 0 - 8 MB               | NPU                  |
| charsiu_encoder | VOICE_AI | float     | Snapdragon® 8 Gen 1 Mobile       | 1.344 ms            | 0 - 9 MB               | NPU                  |
| charsiu_encoder | VOICE_AI | float     | Qualcomm® Dragonwing™            | 2.791 ms            | 0 - 9 MB               | NPU                  |



| Model           | Runtime  | Precision | Chipset                               | Inference Time (ms) | Peak Memory Range (MB) | Primary Compute Unit |
|-----------------|----------|-----------|---------------------------------------|---------------------|------------------------|----------------------|
| charsiu_encoder | VOICE_AI | float     | Snapdragon® 8 Elite Mobile            | 0.527 ms            | 0 - 13 MB              | NPU                  |
| charsiu_encoder | VOICE_AI | float     | Snapdragon® 8 Elite Gen 5 Mobile      | 0.489 ms            | 0 - 9 MB               | NPU                  |
| decoder         | VOICE_AI | float     | Snapdragon® X2 Elite                  | 1.741 ms            | 0 - 0 MB               | NPU                  |
| decoder         | VOICE_AI | float     | Snapdragon® X Elite                   | 3.071 ms            | 0 - 0 MB               | NPU                  |
| decoder         | VOICE_AI | float     | Snapdragon® 8 Gen 3 Mobile            | 2.201 ms            | 0 - 8 MB               | NPU                  |
| decoder         | VOICE_AI | float     | Snapdragon® 8 Gen 1 Mobile            | 4.395 ms            | 0 - 9 MB               | NPU                  |
| decoder         | VOICE_AI | float     | Qualcomm® Dragonwing™ QCS8275         | 8.069 ms            | 0 - 9 MB               | NPU                  |
| decoder         | VOICE_AI | float     | Qualcomm® Dragonwing™ QCS8550 (Proxy) | 3.02 ms             | 0 - 1 MB               | NPU                  |
| decoder         | VOICE_AI | float     | Qualcomm® SA8775P                     | 3.49 ms             | 0 - 10 MB              | NPU                  |
| decoder         | VOICE_AI | float     | Qualcomm® SA8650P                     | 3.49 ms             | 0 - 10 MB              | NPU                  |
| decoder         | VOICE_AI | float     | Qualcomm® SA8255P                     | 3.49 ms             | 0 - 10 MB              | NPU                  |

| Model   | Runtime  | Precision | Chipset                          | Inference Time (ms) | Peak Memory Range (MB) | Primary Compute Unit |
|---------|----------|-----------|----------------------------------|---------------------|------------------------|----------------------|
| decoder | VOICE_AI | float     | Qualcomm® QCS8450                | 4.395 ms            | 0 - 9 MB               | NPU                  |
| decoder | VOICE_AI | float     | Qualcomm® Dragonwing™ IQ-9075    | 3.33 ms             | 0 - 2 MB               | NPU                  |
| decoder | VOICE_AI | float     | Qualcomm® Dragonwing™ IQ-X7181   | 3.071 ms            | 0 - 0 MB               | NPU                  |
| decoder | VOICE_AI | float     | Qualcomm® Dragonwing™ Q-8750     | 1.897 ms            | 0 - 12 MB              | NPU                  |
| decoder | VOICE_AI | float     | Qualcomm® SA7255P                | 8.069 ms            | 0 - 9 MB               | NPU                  |
| decoder | VOICE_AI | float     | Qualcomm® SA8295P                | 3.945 ms            | 0 - 6 MB               | NPU                  |
| decoder | VOICE_AI | float     | Snapdragon® 8 Elite Mobile       | 1.897 ms            | 0 - 12 MB              | NPU                  |
| decoder | VOICE_AI | float     | Snapdragon® 8 Elite Gen 5 Mobile | 1.827 ms            | 0 - 8 MB               | NPU                  |
| encoder | VOICE_AI | float     | Snapdragon® X2 Elite             | 18.626 ms           | 0 - 0 MB               | NPU                  |
| encoder | VOICE_AI | float     | Snapdragon® X Elite              | 29.667 ms           | 0 - 0 MB               | NPU                  |
| encoder | VOICE_AI | float     | Snapdragon® 8 Gen 3 Mobile       | 23.636 ms           | 0 - 7 MB               | NPU                  |



| Model   | Runtime  | Precision | Chipset                               | Inference Time (ms) | Peak Memory Range (MB) | Primary Compute Unit |
|---------|----------|-----------|---------------------------------------|---------------------|------------------------|----------------------|
| encoder | VOICE_AI | float     | Qualcomm® SA7255P                     | 46.989 ms           | 0 - 10 MB              | NPU                  |
| encoder | VOICE_AI | float     | Qualcomm® SA8295P                     | 35.241 ms           | 0 - 6 MB               | NPU                  |
| encoder | VOICE_AI | float     | Snapdragon® 8 Elite Mobile            | 18.973 ms           | 0 - 8 MB               | NPU                  |
| encoder | VOICE_AI | float     | Snapdragon® 8 Elite Gen 5 Mobile      | 17.116 ms           | 0 - 9 MB               | NPU                  |
| flow    | VOICE_AI | float     | Snapdragon® X2 Elite                  | 9.679 ms            | 4 - 4 MB               | NPU                  |
| flow    | VOICE_AI | float     | Snapdragon® X Elite                   | 15.874 ms           | 4 - 4 MB               | NPU                  |
| flow    | VOICE_AI | float     | Snapdragon® 8 Gen 3 Mobile            | 11.076 ms           | 4 - 12 MB              | NPU                  |
| flow    | VOICE_AI | float     | Snapdragon® 8 Gen 1 Mobile            | 18.261 ms           | 4 - 13 MB              | NPU                  |
| flow    | VOICE_AI | float     | Qualcomm® Dragonwing™ QCS8275         | 39.518 ms           | 1 - 10 MB              | NPU                  |
| flow    | VOICE_AI | float     | Qualcomm® Dragonwing™ QCS8550 (Proxy) | 15.032 ms           | 4 - 5 MB               | NPU                  |
| flow    | VOICE_AI | float     | Qualcomm® SA8775P                     | 17.273 ms           | 1 - 10 MB              | NPU                  |

| Model | Runtime  | Precision | Chipset                          | Inference Time (ms) | Peak Memory Range (MB) | Primary Compute Unit |
|-------|----------|-----------|----------------------------------|---------------------|------------------------|----------------------|
| flow  | VOICE_AI | float     | Qualcomm® SA8650P                | 17.273 ms           | 1 - 10 MB              | NPU                  |
| flow  | VOICE_AI | float     | Qualcomm® SA8255P                | 17.273 ms           | 1 - 10 MB              | NPU                  |
| flow  | VOICE_AI | float     | Qualcomm® QCS8450                | 18.261 ms           | 4 - 13 MB              | NPU                  |
| flow  | VOICE_AI | float     | Qualcomm® Dragonwing™ IQ-9075    | 17.415 ms           | 4 - 10 MB              | NPU                  |
| flow  | VOICE_AI | float     | Qualcomm® Dragonwing™ IQ-X7181   | 15.874 ms           | 4 - 4 MB               | NPU                  |
| flow  | VOICE_AI | float     | Qualcomm® Dragonwing™ Q-8750     | 8.942 ms            | 1 - 10 MB              | NPU                  |
| flow  | VOICE_AI | float     | Qualcomm® SA7255P                | 39.518 ms           | 1 - 10 MB              | NPU                  |
| flow  | VOICE_AI | float     | Qualcomm® SA8295P                | 18.898 ms           | 0 - 6 MB               | NPU                  |
| flow  | VOICE_AI | float     | Snapdragon® 8 Elite Mobile       | 8.942 ms            | 1 - 10 MB              | NPU                  |
| flow  | VOICE_AI | float     | Snapdragon® 8 Elite Gen 5 Mobile | 8.975 ms            | 3 - 11 MB              | NPU                  |
| sdp   | VOICE_AI | float     | Snapdragon® X2 Elite             | 7.147 ms            | 0 - 0 MB               | NPU                  |



| Model | Runtime  | Precision | Chipset                          | Inference Time (ms) | Peak Memory Range (MB) | Primary Compute Unit |
|-------|----------|-----------|----------------------------------|---------------------|------------------------|----------------------|
| sdp   | VOICE_AI | float     | Qualcomm® Dragonwing™ IQ-X7181   | 11.333 ms           | 0 - 0 MB               | NPU                  |
| sdp   | VOICE_AI | float     | Qualcomm® Dragonwing™ Q-8750     | 6.956 ms            | 0 - 10 MB              | NPU                  |
| sdp   | VOICE_AI | float     | Qualcomm® SA7255P                | 20.547 ms           | 0 - 9 MB               | NPU                  |
| sdp   | VOICE_AI | float     | Qualcomm® SA8295P                | 12.668 ms           | 0 - 6 MB               | NPU                  |
| sdp   | VOICE_AI | float     | Snapdragon® 8 Elite Mobile       | 6.956 ms            | 0 - 10 MB              | NPU                  |
| sdp   | VOICE_AI | float     | Snapdragon® 8 Elite Gen 5 Mobile | 6.608 ms            | 0 - 9 MB               | NPU                  |

## License

- The license for the original implementation of PiperTTS-DE can be found [here](#).

## References

- [PiperTTS High-quality Multi-lingual Multi-accent Text-to-Speech](#)
- [Source Model Implementation](#)

## Community

- Join [our AI Hub Slack community](#) to collaborate, post questions and learn more about on-device AI.
- For questions or feedback please [reach out to us](#).

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